| Address | Value |
| --- | --- |
| 40H | 05H |
| 41H | 10H |
| 42H | 20H |
| 43H | 25H |
| 44H | 35H |

# Sorting Of Number

AIM:

To write and execute an assembly language program for sorting data in ascending and descending order using 8051 microcontroller on Keil Software.

APPARATUS REQUIRED:

Personal Computer
Keil μ Vision Software

PROGRAM:

```
MOV R7, #4
LOOP1:
MOV R0, #4H
MOV R6, #04
LOOP:
MOV A, @R0
INC R0
MOV 80H, @R0
LJNE A, 80H, NEXT
SJMP DOWN
NEXT: JNC DOWN
MOV @R0, A
DEC R0
MOV @R0, 50H
DOWN: DJNZ R6, LOOP
DJNZ R7, LOOP1
END
```

RESULT:

Thus the sorting of data is Calculated using 8051 in Keil Software.

# 8051 Descending Program:

AIM:

To write and execute an Assembly Language Program for sorting data in descending order using 8051 in Keil Software.

APPARATUS REQUIRED:

Personal Computer

Keil μ Vision Software

PROGRAM:

```
MOV R7, #04H
LOOP 1 : MOV R0, #40H
MOV RL, #04H
LOOP: MOV A, @ R0
INC, R0
MOV 50H, @ R0
LJNE A, 50H, NEXT
SJMP DOWN
NEXT: JL DOWN
MOV @ R0, A
DEC R0
MOV @ R0, 50H
DOWN: DJNZ R6, LOOP
PJNZ RJ, LOOP1
END
```

RESULT:

Thus, the sorting of data in descending order is calculated using 8051 in Keil Software.